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RESEARCH ARTICLE

Real-Time Obstacle Detection and Motion Analysis for Visually Impaired Navigation With Audio Feedback

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ABSTRACT This paper presents a vision-based real-time assistive navigation system that operates exclusively from a single monocular camera, achieving ego-motion compensated dynamic obstacle classification without LiDAR, stereo vision, or specialized hardware. The proposed architecture addresses the fundamental ambiguities of monocular perception through a depth-adaptive multi-cue confidence fusion framework integrating six complementary motion indicators: optical flow, position displacement, appearance similarity, trajectory linearity, size change analysis, and adaptive weighting. The system comprises five modules: Preprocessing, Obstacle Detection, Detection Handling, Movement Detection, and Audio Feedback. The lightweight YOLOv8n architecture (3M parameters) enables real-time operation with an average pipeline latency of 46.73ms, evaluated on a resource-constrained laptop GPU (NVIDIA RTX 3050, 4GB VRAM) to approximate portable hardware conditions. The system identifies six object categories with an overall mAP@0.5 of 87.1%, evaluated on an independent held-out test set of 300 images with zero overlap with training data, maintaining 81.8% under low-light conditions.

INDEX TERMS Machine learning, AI in disability, deep learning, computer vision, assistive technology, visual impairment, obstacle detection, optical flow, motion compensation.

I. INTRODUCTION

Artificial intelligence, and particularly its subfield of computer vision, has demonstrated transformative capabilities across diverse domains — from autonomous vehicle navigation in complex urban environments to early-stage disease detection in medical imaging. Despite these advances and the broad commercial deployment of AI-driven solutions, the application of computer vision to assist persons with disabilities, particularly the visually impaired, remains significantly underexplored.

According to the World Health Organization, approximately 43 million individuals are totally blind and 295 million are moderately to severely sight-impaired globally. This population faces persistent challenges in navigation, object recognition, and environmental awareness — challenges that could be substantially mitigated through strategic application

of modern computer vision technologies. The disparity between rapid AI advancement and its limited deployment in accessibility solutions represents a critical gap in equitable technology distribution.

Traditional assistive technologies such as white canes and guide dogs, while effective for basic obstacle avoidance, are limited in scope and functionality. They fail to provide contextual situational awareness — such as the nature, class, or movement state of nearby objects — which is essential for safe navigation in dynamic urban environments. Furthermore, existing electronic navigation aids often lack real-time motion awareness, ego-motion compensation, or the ability to distinguish between static and dynamic obstacles.

Conversely, the computer vision systems of the modern time promise the possibility of analysing the environment in real time, recognizing objects, and monitoring dynamic obstacles, thereby making it possible to have a more detailed situational awareness. These systems can be utilized to fill the gap between the old-fashioned assistance and the increasing

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demand of smart, situation-aware navigation help in dynamic environments, using advanced sensing technologies and machine learning algorithms.

This research presents a practical and accessible assistive solution that operates exclusively from a single monocular camera without LiDAR, stereo vision, ultrasonic sensors, or specialized hardware. A single monocular camera presents two fundamental perceptual challenges: it cannot directly measure depth, and it cannot inherently distinguish camera motion from object motion. These ambiguities are the core problem that the proposed system is designed to solve. By leveraging YOLOv8 [5], [32] for lightweight object detection and a novel depth-adaptive multi-cue confidence fusion framework for motion classification, the system achieves ego-motion compensated dynamic obstacle detection in real-time using Optical Flow [12]. Every architectural component, the heuristic depth estimation, homography based ego motion compensation, six cue adaptive weighting, and asymmetric hysteresis thresholds exists specifically to overcome the inherent constraints of single-camera monocular perception.

The main contributions of this work are:

- We propose a real time outdoor navigation assistive framework that operates exclusively from a single camera, integrating computer vision and visual auditory substitution to support safe mobility for visually impaired users, without requiring LiDAR, depth sensors, stereo vision, or specialized hardware.
- We propose a depth adaptive multi cue confidence fusion framework that dynamically reweights six complementary motion indicators — optical flow, position displacement, appearance similarity, trajectory linearity, size change, and adaptive weighting based on estimated object depth. This addresses a fundamental limitation of monocular vision: the reliability of individual motion cues varies significantly with object distance, and a fixed weight fusion cannot account for this variation.
- We develop a low-latency processing pipeline achieving 46.73ms average latency, evaluated on a resource constrained testing environment deliberately selected to approximate embedded hardware conditions, with YOLOv8n's ~3M parameter count supporting feasibility for future deployment on platforms such as NVIDIA Jetson Nano or Raspberry Pi.

The remainder of this paper is organized as follows: Section II related works, Section III details the proposed methodology, Section IV outlines the experimental results, Section V describes comparison, Section VI gives discussion, Section VII concludes the paper, and Section VIII discusses the future work.

II. RELATED WORKS

Navigation problems continue to be one of the greatest challenges towards mobility of people with visual impairment. The dynamic environment can be investigated in real-time,

though not many studies have been conducted on how this can be done despite the evolution of the field of assistive technologies. Several other past researches have explored the use of aids, and the application of computer vision, sensor fusion and portability.

Krishna et al. [19] proposed a system using 3D audio spatial feedback to convey obstacle location and direction, enhancing spatial perception. However, the need for complex spatial mapping and audio calibration limits its portability and responsiveness. Gautam et al. [13] focused on vision-based object grasping, which facilitates short-range tasks but lacks adaptability for dynamic outdoor navigation.

Dimas et al. [8] introduced a self-supervised method for detecting soft obstacles by analyzing visual patterns in video data. While their learning strategy is flexible, the system lacks real-time inference and feedback. Similarly, the Envision system [24] supports digital interaction but does not address real-world dynamic obstacle avoidance. Other systems such as Walking Assistant [1] and Kommey et al.'s smart vision aid [23] assist in general navigation but overlook motion estimation and dynamic object prioritization.

Dubey et al. [10] applied YOLO with optical flow to provide voice feedback on nearby objects, yet they omit multi-object tracking and motion differentiation, making them less effective in unstructured settings. Wang et al. [30] presented Motor Focus, an ego-motion prediction model, which is well-suited for motion analysis but does not include object interaction or user feedback. Similarly, Xie et al. proposed a YOLO and ORB-SLAM integration for indoor mobility, though SLAM-based systems often suffer from computational overhead and reduced outdoor usability.

Srinivasiah et al. [28] proposed a sensor fusion framework for visually impaired navigation employing extended Kalman filtering for homogeneous sensor fusion and rule-based fusion for heterogeneous sensors, though its complexity limits deployment on lightweight portable hardware. Likewise, Busaeed [6] developed LidSonic V2.0, a low-cost wearable system using LiDAR and edge computing for efficient obstacle classification. While energy-efficient, LidSonic lacks visual detail and motion-awareness capabilities. Liu et al. [22] proposed HIDA, which uses LiDAR-based semantic segmentation for indoor understanding, but it does not generalize well to dynamic outdoor scenarios.

Jemin and Sasikumar [15] proposed ECEHLY-V2, combining a lightweight YOLO-fastest V2 model with an improved elk herd optimizer to enhance obstacle detection and tracking, reporting accuracy at 99.8%, precision at 98.65%, recall at 98.25%, and F1-score at 97.64%. Despite strong detection performance, the framework focuses primarily on obstacle detection and tracking, with no reported mechanism for ego-motion compensation or dynamic obstacle classification.

Wang et al. [31] introduced YOLO-OD, a YOLOv8-based obstacle detection model incorporating three novel modules — a Feature Weighting Block (FWB), an Adaptive

Bottleneck Block (ABB), and an Enhanced Feature Attention Head (EFAH) — achieving a mAP of 30.02% on a public road obstacle dataset. However, the proposed method focuses on obstacle detection rather than obstacle motion-state analysis and does not report mechanisms for motion compensation or dynamic obstacle classification.

Bastidas-Guacho et al. [3] presented a mobile computer vision system integrating object detection with multimodal auditory and haptic feedback for visually impaired individuals. Their system was evaluated directly with impaired users, demonstrating practical accessibility benefits; however, the best-performing YOLO-300 model achieved a mAP of 0.42, and the system lacks motion-aware processing or ego-motion compensation.

Das et al. [7] proposed PathFinder, a map-less navigation system combining Vision Language Models (VLMs), Large Language Models (LLMs), and monocular depth estimation with a Depth-First Search algorithm for free-path detection in both indoor and outdoor environments. While PathFinder demonstrates strong usability with blind and low-vision participants, it depends on computationally intensive language models.

Farid et al. [11] proposed an assistive navigation system that transforms monocular visual information into spatial audio cues for visually impaired users. The framework combines computer vision and real-time audio rendering techniques within an embedded deployment setting, enabling navigation assistance without requiring stereo cameras or additional ranging sensors. Although the system provides intuitive spatial audio feedback, the reported methodology does not explicitly address ego-motion compensation or dynamic obstacle motion-state classification.

Joshi et al. [17] proposed AI-SenseVision, a low-cost wearable assistive system that combines artificial intelligence techniques with sensor-based obstacle avoidance to support navigation for visually impaired people. The framework provides auditory guidance based on detected objects and obstacles and was evaluated in real-world navigation scenarios. While the system demonstrates effective integration of computer vision and sensing technologies, the reported methodology relies on dedicated ranging sensors and does not explicitly address motion compensation or dynamic obstacle state classification.

Noor et al. [2] conducted a feasibility study for a wearable IoT-based indoor object detection system using a Raspberry Pi 4 and YOLOv11, evaluated on a furniture dataset containing 3000 images across three object categories. The system achieved 99.07%, 98.51%, 97.96%, and 98.22% for accuracy, precision, recall, and F1-score, respectively. While the study demonstrates the feasibility of real-time object detection on embedded hardware, its evaluation is limited to indoor furniture recognition using a small object taxonomy and does not investigate motion-aware navigation, dynamic obstacle analysis, or outdoor deployment scenarios.

More advanced vision-centric models such as PC-CS-YOLO [21] achieve high detection precision through partial

convolution and attention fusion. However, they focus on static object detection and omit motion cues critical to mobile navigation. MR.NAVI [25] is a system which introduces a mixed-reality interface combining visual overlays with scene understanding but depends on MR headsets and does not support motion-based segmentation or lightweight deployment.

The recent developments in assistive technologies have brought about a massive improvement in the independence of the visually impaired people by incorporating the computer vision and sensory substitution techniques. The outdoor navigation assistive system proposed by Scalvini et al. creates a real-time visual auditory substitution model that transforms visual scene detail into intuitive auditory signals to enhance the spatial cognition and safe navigation in the outdoor setting. The system allows users to recognize obstacles and comprehend surrounding environments effectively by adding a feature of real-time object detection, depth estimation and spatialized audio feedback. The possibilities of real-time multimodal technologies in enhancing the autonomy and safety of the visually impaired users in dynamic outdoor situations are demonstrated by such intelligent assistive systems [26] such as these can show the potential of real-time multimodal technologies to improve the autonomy and safety of the intended users who are visually impaired in dynamic outdoor settings.

Systems achieving motion awareness or depth perception typically depend on additional hardware — LiDAR [6], [22], depth cameras [19], or mixed-reality headsets [25]. Silva and Wimalaratne [28] require multi-sensor fusion with extended Kalman filtering. LidSonic V2.0 [6] requires dedicated LiDAR hardware. HIDA [22] requires a wearable solid state LiDAR sensor. These hardware requirements increase system cost, weight, and calibration complexity, directly limiting accessibility and portability for the target population. To our knowledge, no prior assistive navigation system achieves ego-motion compensated dynamic obstacle classification with depth-adaptive multi-cue motion fusion using only a single monocular camera without additional sensors.

Transformer-based architectures [9] rely on multi-head self-attention mechanisms whose computational cost scales with sequence length and embedding dimension. At a processing resolution of 1280×720 with a patch size of 16×16 , this produces $n = 3,600$ tokens per frame; for ViT-B/16 ($\sim 86M$ parameters, 12 attention layers), the resulting attention cost per frame exceeds practical embedded budgets. Similarly, Mamba-based state space models [14] depend on CUDA-specific selective scan kernels that are unavailable on the ARM Cortex-A57 (Jetson Nano) and ARM Cortex-A76 (Raspberry Pi 5) processor architectures, preventing CPU fallback execution. In contrast, the proposed system utilizes YOLOv8n with its compact 3M parameter count and 6 MB model, fewer parameters than ViT-B/16 where only the detection inference requires GPU. The entire motion analysis pipeline, including homography estimation, optical

TABLE 1. Comparison of related vision-based assistive systems.

System	Detection Method	Motion Awareness	Feedback	Real-Time	Advantages	Limitations
Krishna et al. [24]	CNN-based object detection with stereoscopic depth sensing	Limited (3D localization only)	Spatial 3D audio via earphones	Partial	Intuitive spatial audio feedback; natural object positioning using depth map	Requires stereo camera setup; limited to door detection; no dynamic obstacle tracking
Gautam et al. [16]	Vision-based object detection for grasping	No	Audio guidance	No	Effective for close-range object grasping tasks	No outdoor navigation; no motion estimation; not real-time
Dimas et al. [1]	Self-supervised soft obstacle detection from video patterns	No	None	No	No manual annotation required; flexible self-supervised learning	No real-time inference; no user feedback; not validated outdoors
Envision System [23]	Mobile application with vision-based scene detection	No	Text-to-speech	Yes (not navigation focused)	Accessible smartphone deployment; broad scene description	Not designed for obstacle avoidance or navigation; no spatial awareness
YOLO Tracking [18]	YOLO with multi-object tracking	Yes (via tracking)	Visual monitoring only	Yes	Real-time multi-object tracking; established detection pipeline	No audio feedback for visually impaired; no ego-motion compensation
Our Proposed System	YOLOv8n with Lucas-Kanade optical flow and RANSAC homography	Yes (ego-motion aware)	Audio (eSpeak) with spatial zones	Yes (46.73ms)	Lightweight; ego-motion compensation; no extra hardware; 87.1% mAP@0.5 on independent test set	Performance degrades on signal class under low-light conditions

flow, SSIM computation, and trajectory filtering, executes on CPU, making the system directly compatible with embedded platforms.

Conversely, the proposed solution provides a system that integrates object detection with the use of the YOLO algorithm and dense optical flow and template matching algorithms to track a trajectory. It is motion aware and can be used to give real time feedback without the use of other hardware like LiDAR or ultrasonic sensors. This makes it possible to distinguish between possessing a static and dynamic object, to prioritize obstacles effectively, and to be able to be deployed within unstructured outdoor settings that mitigates the main limitations present in both vision-only and sensor-fusion-based prior work.

III. METHODOLOGY

The proposed system comprises five major components: (A) preprocessing, (B) obstacle detection, (C) detection handling, (D) movement detection, and (E) feedback.

A. PRE-PROCESSING

The main task of this module is to equip the input information to the obstacle detection model. It does a set of pre-processing to make it compatible with the input requirement of the model. It starts by extracting frames in which the input video is broken down to separate frames. The frames are then scaled to 1280×720 pixels then converted to grayscale. The processed frames are now sent to the next module.

B. OBSTACLE DETECTION

This module has the duty of identifying and assigning labels to the obstacles found in each frame. As such, we are using the YOLOv8n model, which is a lightweight and

TABLE 2. Object classifications.

No	Classification
1	Crosswalks
2	Pedestrians
3	Vehicles
4	Generic Obstacles
5	Traffic Signals
6	Stairs

pre-trained object detection model that can be modified to fit a particular application. In our scenario, a model is trained on the identification of obstacles with the help of a **custom dataset of 7,083** training images across six classes. The dataset was split using a **90/10 train/validation ratio**, with an independent held-out **test set of 600 images** created to represent 2 conditions prior to any training to ensure zero overlap with training data. The model is trained to learn and recognize six categories:

C. DETECTION HANDLING

This module processes the raw outputs from the object detection stage. Each detected object is associated with a confidence score and a class label. Low-confidence detections are discarded to reduce false positives. The remaining objects are stored with their bounding box coordinates, object class, and timestamp for temporal tracking in the movement detection module.

D. MOVEMENT DETECTION

The movement detection module represents the core novel contribution of this work. Operating from a single monocular camera, the system faces two fundamental perceptual challenges that do not exist in sensor fusion systems:

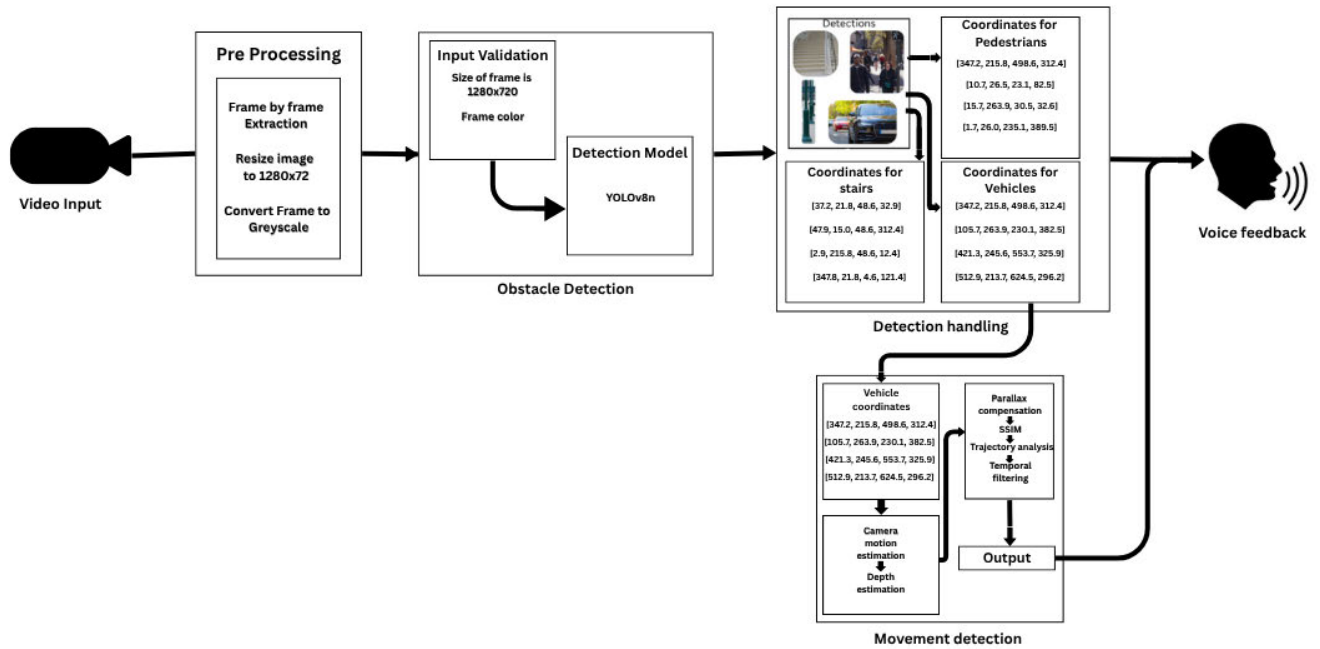


FIGURE 1. Methodology diagram.

(1) depth cannot be directly measured, requiring heuristic estimation from visual geometry; and (2) camera motion cannot be separated from object motion through sensor readings, requiring computational ego motion compensation. Every component of this module — the heuristic depth estimator, homography-based camera motion estimation, six-cue confidence framework, depth-adaptive weighting, and asymmetric hysteresis filter — is a direct architectural response to these monocular constraints.

This module decides whether a vehicle that is detected is dynamic or static. The coordinates of the bounding box calculated at the Detection Handling module are initially analyzed to determine whether the object identified is of the vehicle category or not. In case of confirmation, then the respective region is sent to the Movement Detection module to undergo analysis.

To address the problem of identifying truly moving vehicles and not motion due to camera motion our movement detector system uses multi modality technique. This starts by estimating the motion of camera through homography transformation. Background feature points are identified through the Shi-Tomasi corner detector [27] (with quality level 0.01 and up to 2000 points), explicitly masking out vehicle regions with a 10-pixel buffer.

These points are tracked between consecutive frames using Lucas-Kanade optical flow with a 21×21 window size, followed by RANSAC-based homography calculation with a 3.0-pixel threshold.

$$\frac{\partial I}{\partial x} \cdot v_x + \frac{\partial I}{\partial y} \cdot v_y + \frac{\partial I}{\partial t} = 0$$

$$\|H \cdot p_1 - p_2\| < \text{threshold}$$

The homography matrix (H) resulting is used to parameterize the motion of the entire camera, and this has been verified by inlier ratio and motion magnitude values and p1 and p2 depict points of frames.

On strategies with each tracked vehicle, we use parallax compensation on depth estimation. The computation of the relative depth is computed as a weighted average of the size cues and vertical position along the formula.

$$0.6 \times (1 - \min(1.0, AR \times 30)) + 0.4 \times (1 - BP)$$

where AR denotes the area ratio and BP denotes the bottom position.

The coefficient 0.6 denotes **SIZE_WEIGHT**, 0.4 denotes **POSITION_WEIGHT**, and the scalar 30 denotes **SIZE_FACTOR**.

SIZE_WEIGHT and **POSITION_WEIGHT** define the relative contribution of the size cue and vertical position cue respectively, satisfying **SIZE_WEIGHT** + **POSITION_WEIGHT** = 1.0.

SIZE_FACTOR controls the sensitivity of the area-based depth cue. **MIN_DEP** and **MAX_DEP** in the CLAMP operation bound the output to a normalized range, preventing degenerate depth values at extreme object scales.

The Detection Module includes additional features designed to reduce false detections. It Implement two distinct optical flow computations which serves the system different purposes: sparse Lucas-Kanade optical flow is used exclusively for background feature point tracking to estimate camera motion via homography, while dense optical flow is computed independently within each vehicle's bounding box

Algorithm 1 Depth Estimation

Require: bounding_box = $[x_1, y_1, x_2, y_2]$,
frame_dimensions = $[W, H]$

Ensure: estimated_depth $\in [0.0, 1.0]$

- 1: **Constants:** SIZE_FACTOR = 30, SIZE_WEIGHT = 0.6,
POSITION_WEIGHT = 0.4,
MIN_DEP = 0.0, MAX_DEP = 1.0
- 2: width $\leftarrow x_2 - x_1$ ▷ Object width
- 3: height $\leftarrow y_2 - y_1$ ▷ Object height
- 4: object_area $\leftarrow \text{width} \times \text{height}$
- 5: frame_area $\leftarrow W \times H$
- 6: area_ratio $\leftarrow \text{object_area} / \text{frame_area}$
- 7: size_cue $\leftarrow 1 - \min(1.0, \text{area_ratio} \times \text{SIZE_FACTOR})$
- 8: bottom_position $\leftarrow y_2 / H$
- 9: position_cue $\leftarrow 1 - \text{bottom_position}$
- 10: estimated_depth $\leftarrow \text{SIZE_WEIGHT} \times \text{size_cue} + \text{POSITION_WEIGHT} \times \text{position_cue}$
- 11: estimated_depth $\leftarrow \text{CLAMP}(\text{estimated_depth}, \text{MIN_DEP}, \text{MAX_DEP})$

return estimated_depth ▷ Final depth estimate

region for motion classification. These are separate pipeline stages and should not be conflated.

1) DIRECT OPTICAL FLOW ANALYSIS

This method performs dense optical flow computation directly within each vehicle's bounding box region, providing the most reliable motion indicator for close-range objects.

$$\text{FC} = 0.7 \times \min\left(1.0, \frac{\text{mean_magnitude}}{\text{magnitude_threshold}}\right) + 0.3 \times \text{consistency_factor} \quad (1)$$

FC = flow confidence; mean_magnitude is the spatial average of optical flow vector magnitudes within the bounding box region; magnitude_threshold is the depth-adaptive value from Table 3; and consistency_factor $\in [0, 1]$ is the fraction of flow vectors whose orientation lies within 30° of the dominant flow direction, measuring spatial coherence of the motion field.

The magnitude threshold is depth-adaptive, with higher thresholds for closer objects to prevent false positives from camera motion as shown in Table 3.

TABLE 3. Depth-adaptive thresholds for optical flow analysis.

Depth Category	Magnitude Threshold	Consistency Threshold
Extremely Close ($d < 0.1$)	1.5	0.6
Close ($0.1 \leq d < 0.3$)	1.1	0.5
Medium ($0.3 \leq d \leq 0.6$)	0.7	0.35
Distant ($d > 0.6$)	0.5	0.3

2) POSITION-BASED ANALYSIS

The technique determines the distance between the actual and the desired movement of the object caused by the camera.

The adaptive threshold scaling process modulates sensitivity to movement according to the size of the object and the approximate depth:

$$\text{position_threshold} = \max(1.0, \text{diagonal} \times \text{sensitivity_factor}(d)) \quad (2)$$

For extremely close objects ($d < 0.1$), a minimum movement requirement prevents small vibrations from registering as movement:

$$\text{PC} = \begin{cases} 0.1 & \text{if } M < 3.5 \text{ pixels and } d < 0.1 \\ \min\left(1.0, \frac{M}{\text{PT}}\right) & \text{otherwise} \end{cases} \quad (3)$$

PC = Position confidence

M = Movement

PT = Position threshold

Diagonal d is the Euclidean length of the bounding box diagonal in pixels. sensitivity_factor(d) is a depth-dependent scaling coefficient that modulates how much displacement is required to register as motion. The value 3.5 pixels represents the empirically determined threshold below which displacements are attributed to sensor noise rather than true object motion.

3) APPEARANCE-BASED ANALYSIS

In this method, Structural Similarity Index Measure (SSIM) is utilized to perceive movement between frames by comparison of the visual appearance of the tracked objects. The system uses depth-adaptive SSIM thresholds: closer objects have higher thresholds:

$$\text{AC} = \max\left(0, \min\left(1, \frac{\text{ssim_threshold}(d) - \text{ssim_score}}{\text{ssim_threshold}(d)}\right)\right) \quad (4)$$

AC = appearance confidence

ssim_score $\in [-1, 1]$ is the Structural Similarity Index Measure computed between grayscale crops of the object bounding box in the current and previous frames, resized to a common resolution with a minimum of 10×10 pixels. ssim_threshold(d) is a depth-adaptive threshold: a higher value is applied to close-range objects where camera vibration produces measurable appearance change even for static objects, and a lower value is used for distant objects where pixel-level changes are smaller even during genuine motion.

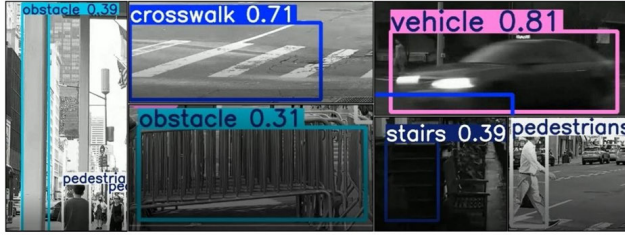
4) TRAJECTORY ANALYSIS

This temporal technique examines the movement of the object over several frames (at least five) in order to identify similar movement patterns. The coefficient of determination (R^2) assesses the linearity of the trajectory:

$$\text{TC} = 0.7 \times \min(1, \max(0, R^2)) + 0.3 \times \min\left(1, \frac{\text{velocity}}{\text{threshold}}\right) + \text{depth_boost} \quad (5)$$

TABLE 4. Depth-adaptive weight allocation for detection methods.

Depth Category	Flow	Position	Trajectory	Appearance	Size
Extremely Close ($d < 0.1$)	0.40	0.25	0.20	0.10	0.05
Close ($0.1 \leq d < 0.3$)	0.35	0.30	0.20	0.10	0.05
Medium ($0.3 \leq d \leq 0.6$)	0.30	0.25	0.30	0.10	0.05
Distant ($d > 0.6$)	0.25	0.20	0.35	0.10	0.10

**FIGURE 2.** Object detection.

TC = trajectory confidence

R^2 is the coefficient of determination from linear regression over the last $N \geq 5$ ego-motion-compensated centroid positions; velocity is the magnitude of the fitted trajectory slope in pixels per frame; and threshold is a depth- and size-dependent reference velocity value that scales with object size ratio for close objects and reduces to a small fixed value for distant objects to increase detection sensitivity. A confidence boost is applied for medium-range objects showing consistent movement:

$$\text{depth_boost} = \begin{cases} 0.12 \times \min(1, R^2) & \text{if } 0.3 \leq d \leq 0.6 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

5) SIZE CHANGE ANALYSIS

This method detects approaching or receding vehicles by analyzing object size variations across frames:

$$\text{size_change_rate} = \frac{|\text{current_area} - \text{previous_area}|}{\max(\text{previous_area}, 1)} \quad (7)$$

6) ADAPTIVE WEIGHTING SYSTEM

The system assigns object depth dynamic weights to each method of detection. Regarding very close objects, the weight given to optical flow is high (0.40) and to the distant objects, the weight given to trajectory analysis is high (0.35).

The combined confidence is calculated as:

$$\text{combined_confidence (CC)} = \sum_{i=1}^5 w_i(d) \times \text{confidence}_i \quad (8)$$

For extremely close objects, an additional confidence penalty is applied to prevent false positives:

$$\text{final_confidence} = \begin{cases} \text{CC} \times 0.7 & \text{if } d < 0.1 \\ \text{CC} & \text{otherwise} \end{cases} \quad (9)$$

This confidence penalty addresses a specific failure mode of monocular optical flow: at very close range, camera motion produces large apparent flow vectors that are easily misclassified as object motion. This penalty is applied only at close range ($d < 0.1$) to avoid suppressing legitimate motion detections at medium and far distances. This depth-specific penalty is a novel design decision not present in prior optical flow based assistive systems [10], [30].

7) ENHANCED TEMPORAL CONSISTENCY FILTERING

This final classification step uses an advanced hysteresis-based filtering mechanism with depth-adaptive parameters. The system maintains state history and applies weighted averaging with recency bias:

$$\text{weighted_avg} = \frac{\sum_{i=1}^n w_i \times \text{state}_i \times \text{confidence}_i}{\sum_{i=1}^n w_i \times \text{confidence}_i} \quad (10)$$

The system implements depth-specific dual thresholds, with extremely close objects requiring much higher evidence (threshold = 0.9) to be classified as moving, while moving objects can transition to static more easily (threshold = 0.15) shown in Table 6.

The asymmetry between static-to-moving thresholds (0.9 for extremely close objects) and moving-to-static thresholds (0.15) is a deliberate safety-motivated design decision specific to assistive navigation. The cost of missing a moving vehicle detection is significantly higher than the cost of a false moving vehicle alert — a false alert causes momentary unnecessary caution, while a missed detection could result in a collision. The asymmetric thresholds encode this safety logic directly into the temporal consistency filter, requiring strong accumulated evidence before classifying an object as newly moving, while readily reclassifying a moving object as static once motion ceases. This asymmetric hysteresis design is not present in prior motion-aware assistive navigation systems.

The parallax factor values reflect the depth-dependent relationship between camera translation and apparent object displacement. Extremely close objects ($d < 0.1$) use a factor of 0.95, meaning nearly all estimated camera motion is subtracted. As depth increases this decreases: close objects use 0.85, medium objects use 0.75, and distant objects use a linearly decaying factor $1.0 - (0.3 \times d)$. When homography H is available, the object centroid is projected through H to obtain the precise camera-induced displacement; otherwise global translation components (dx , dy) are used as fallback.

TABLE 5. Spoken feedback configuration by object type and priority.

Detected Object	Position	Spoken Feedback	Priority Level	Cooldown Period	Conditional Rules
Vehicle	Any	"Vehicle at [position]"	High	None (always announced)	Only one vehicle announcement per frame regardless of count
Stairs	Any	"Stairs at [position]"	High	None (always announced)	Only one stairs announcement per frame
Obstacle	Any	"Obstacle at [position]"	Medium	3 seconds	Only one obstacle announcement per frame
Pedestrian	Any	"Pedestrian at [position]"	Medium	3 seconds	All pedestrian counts treated equally; only one announcement per frame
Crosswalk	Any	"Crosswalk at [position]"	High	3 seconds	Only one crosswalk announcement per frame
Signal	Any	"Proceed to cross"	Medium	3 seconds	Announced only if crosswalk is also detected in the frame; only one signal announcement per frame
Moving vehicle	Any	"Moving vehicle at [position]"	High	None (always announced)	Movement state takes precedence over static detection
Static vehicle	Any	<i>Standard position announcement</i>	High	None (always announced)	Only announced if no moving vehicles of the same class are detected

TABLE 6. Depth-adaptive temporal consistency thresholds.

Depth Category	Static to Moving	Moving to Static
Extremely Close ($d < 0.1$)	0.9	0.15
Close ($0.1 \leq d < 0.3$)	0.85	0.20
Medium ($0.3 \leq d \leq 0.6$)	0.7	0.25
Distant ($d > 0.6$)	0.8	0.20

Algorithm 2 Parallax Compensation**Require:** track_data, camera_motion, estimated_depth**Ensure:** comp_dx, comp_dy: compensated motion vectors

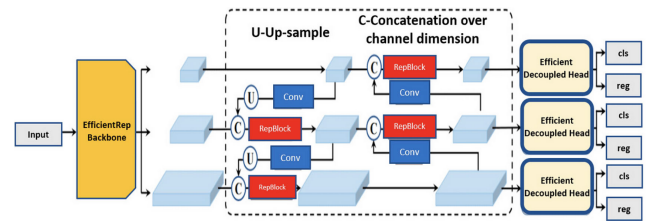
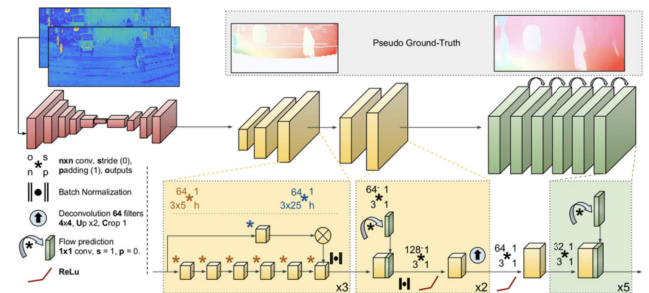
```

1: parallax_factor ← CalculateParallaxFactor(estimated_depth)
2: function CalculateParallaxFactor(d)
3:   if  $d < 0.1$  then
4:     return 0.95           ▷ Extremely close objects
5:   else if  $d < 0.3$  then
6:     return 0.85           ▷ Close objects
7:   else if  $d \leq 0.6$  then
8:     return 0.75           ▷ Medium objects
9:   else
10:    return  $1.0 - (0.3 \times d)$  ▷ Distant objects
11:  end if
12: end function
13: if camera_motion.homography  $\neq$  NULL then
14:   Apply homography transformation with parallax_factor
15: else
16:   Apply simple translation with parallax_factor
17: end if
18: return comp_dx, comp_dy

```

E. FEEDBACK

The Feedback System module serves as the final stage, responsible for voice feedback to the user. This module

**FIGURE 3.** YOLOv8 object detection architecture illustrating backbone feature extraction, multi-scale feature fusion, and decoupled detection heads for outdoor navigation systems [5].**FIGURE 4.** Optical flow architecture illustrating feature extraction and multi-scale convolutional layers for motion estimation and navigation assistance [12].

transforms the analytical outputs from preceding stages into interpretable audio elements using the lightweight eSpeak speech synthesis engine, selected for its minimal computational footprint.

The system utilizes a spatial-temporal feedback architecture whose main elements include position-based categorization, class specific prioritization and temporal filtering. The objects that are detected are placed spatially in one of three positional areas (left, center, right) according to the centroid coordinates of the object compared to the predetermined screen borders.

Algorithm 3 Feedback System for Assistive Navigation

Require: detected_objects, pedestrian_signal, crosswalk_status, announcement_history

Ensure: Voice feedback via eSpeak

```

1: Initialize cooldown_interval, announcement_history
2: for each obj in detected_objects do
3:   zone ← ClassifyPosition(obj.centroid)
4:   if obj.class = "moving vehicle"
   and not RecentlyAnnounced(obj, announcement_history) then
5:     Announce(obj.class, zone)
6:     UpdateAnnouncementHistory(obj)
7:   else if CooldownExpired(obj.class, zone, announcement_history) then
8:     Announce(obj.class, zone)
9:     ResetCooldownTimer(obj.class, zone)
10:    UpdateAnnouncementHistory(obj)
11:   end if
12: end for
13: if CountObjectsOfClass("pedestrian") ≥ 5
   and not RecentlyAnnounced("crowd ahead", announcement_history) then
14:   Announce("crowd ahead")
15:   UpdateAnnouncementHistory("crowd ahead")
16: end if
17: if pedestrian_signal safe and crosswalk_status detected and not
   RecentlyAnnounced("proceed to cross", announcement_history) then
18:   Announce("proceed to cross")
19:   UpdateAnnouncementHistory("proceed to cross")
20: end if
21: CleanOldEntries(announcement_history, hysteresis_time_threshold)

```

The module uses differential announcement schemes (by object class) with priority of moving vehicles being immediately verbalized irrespective of frequency, and common objects being susceptible to cooldown-based filtering (waiting 3 seconds between repeated announcements).

A specialized crowd detection mechanism aggregates multiple pedestrian instances (greater than 5) into a single "crowd ahead" announcement. The system detects pedestrian signals and crosswalks, providing a simple 'proceed to cross' announcement only when three conditions are simultaneously satisfied: a crosswalk is detected in the frame, a pedestrian signal is detected above the system's minimum confidence threshold, and the signal class corresponds to a safe crossing state. This condition gating prevents false crossing announcements in low light or occluded conditions where signal detection confidence is degraded.

The system maintains a chronological record of announcements to implement hysteresis in the feedback loop, thus ensuring stable, non-repetitive audio outputs even in visually

complex environments. Algorithm 3 shows the algorithm for feedback and table 5 shows the feedback for each detected class.

IV. EXPERIMENTAL SETUP

Experiments were conducted on an NVIDIA GeForce RTX 3050 Laptop GPU with an effective available VRAM of 3.5GB intentionally selected as a resource-restricted environment to approximate the computational budget of portable hardware rather than a high-end workstation GPU. The 3.5GB effective VRAM constraint represent conditions significantly more restrictive than desktop GPUs, providing a conservative performance estimate relevant to embedded deployment. Deployment on platforms such as the NVIDIA Jetson Nano and Raspberry Pi 5 is identified as an immediate priority for future work.

Training Configuration:

- Model: YOLOv8n
- Batch size: 8
- Epochs: 150 (early stopping at epoch 82, best model at epoch 52)
- Dataset: Custom, self made (7,083 train / 765 val / 300 test)
- Initial learning rate (lr_0): 0.01
- Final learning rate (lr_f): 0.001
- Optimizer: AdamW
- Total parameters: 3,006,818 (~3.0M)

A. DATASET USED

A custom dataset was constructed for six-class urban pedestrian navigation, comprising the following categories: crosswalk, pedestrian, vehicle, obstacle, traffic signal, and stairs. No existing public benchmark - including COCO or ImageNet - provides a direct mapping to this taxonomy, which requires domain-specific annotation expertise, careful selection of urban environments relevant to visually impaired users, and deliberate handling of class imbalance across safety-critical categories. **Starting from 1,142 manually annotated images, targeted augmentation expanded the training corpus to 7,083 images.** Augmentation techniques applied include horizontal flipping, rain effect simulation via brightness reduction combined with diagonal streak overlay, low-light simulation through brightness reduction, and zoom augmentation. The signal and stairs classes, originally containing 480 and 301 images respectively, received additional targeted augmentation rounds to mitigate severe class imbalance. A 90/10 train-validation split was applied. An independent held-out test set of 300 standard-lighting images was constructed prior to any training activity to ensure zero data leakage. An additional 300 low-light test images were generated by applying brightness reduction to a separate image sample, enabling robustness evaluation under adverse illumination conditions. The complete evaluation set therefore comprises 600 images across two distinct lighting conditions, with no overlap with training data. The

TABLE 7. Latency breakdown of the motion analysis pipeline.

Component	Mean (ms)	Std. Dev. (ms)	% of Total time taken	Function
Camera Motion Estimation	39.87	12.84	85.3%	Shi–Tomasi feature extraction with Lucas–Kanade optical flow and RANSAC-based homography for ego-motion compensation
Track Data Management	3.42	2.12	7.3%	Object tracking state updates and historical data maintenance
Appearance Similarity (SSIM)	3.19	2.18	6.8%	Frame-to-frame visual consistency analysis for motion detection
Trajectory Analysis	0.20	0.03	0.4%	Multi-frame position tracking for sustained motion pattern analysis
Temporal Filtering	0.03	0.00	0.1%	Hysteresis-based temporal smoothing for motion classification
Other Components	0.02	0.01	0.1%	Position analysis, size variation handling, adaptive weighting, and parallax compensation
Total Pipeline	46.73	15.64	100%	Complete detection and motion analysis pipeline

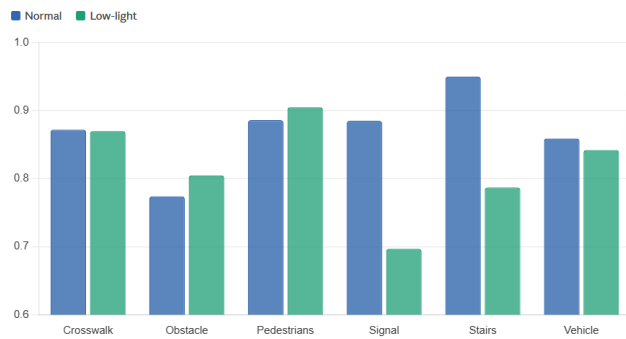


FIGURE 5. Class wise result and Low light scenario result.

development of a large-scale, curated, and publicly available benchmark for urban pedestrian navigation is identified as a significant open research contribution and is planned as a dedicated future effort.

B. CLASS-WISE DETECTION PERFORMANCE

Detection performance was evaluated using standard object detection metrics, including **Precision**, **Recall**, **mAP@0.5**, and **mAP@0.5:0.95**. Performance was evaluated on an independent test set of 300 images never seen during training. Graph 5 presents a comparison between the model results on test data and on low light scenarios.

Overall mAP@0.5 is 87.1%. Under low-light conditions, the system maintains 81.8% overall mAP@0.5, representing only a 5.3% degradation. Signal detection shows the most significant degradation under low-light conditions 88.5% to 69.7%, attributed to the small bounding box size of traffic signals combined with reduced visual contrast in dark environments.

C. PROCESSING PIPELINE LATENCY

The table 7 shows per-frame processing time including:

- Object detection (YOLO)
- Camera motion estimation
- Multi-cue motion analysis
- Track management

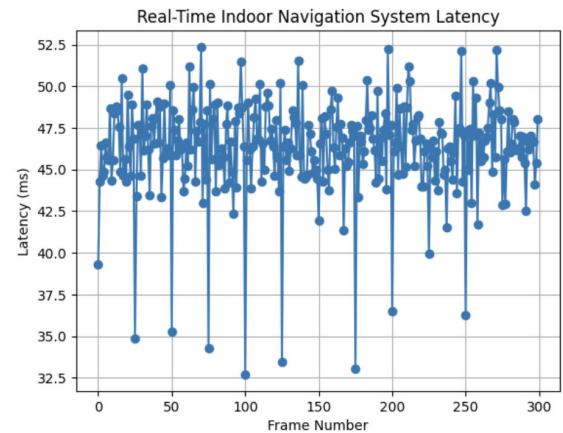


FIGURE 6. Real-time latency per frame of the navigation system showing stable performance with an average latency of 46.73 ms.

The table 7 shows result obtained when the system was run on multiple videos of an average frame count of 200. Combining the YOLOv8n detection inference (4.2ms) with the motion analysis pipeline (46.73), the total latency of the complete system is approximately 50.93ms.

V. COMPARISON

A wide range of technologies have been explored in the development of assistive navigation systems for visually impaired users. Object detection frameworks such as YOLO, particularly its lightweight variant YOLOv8n, have been adopted due to their real-time performance capabilities. The proposed system pairs YOLOv8n and optical flow algorithms such as Lucas-Kanade a differential technique in computer vision for estimating sparse optical flow, tracking feature points (like corners) across video frames by assuming motion is constant in small neighborhoods, and RANSAC based homography for motion compensation which is a robust technique for aligning two images by estimating a 3×3 transformation matrix (homography) while filtering out false feature matches.

Other detection models, including SSD and Faster R-CNN, are also frequently employed, each offering different trade offs between speed and accuracy.

TABLE 8. Model comparison.

Model	Parameters	mAP50	mAP50-95	Inference (ms)
YOLOv5n	2.5M	0.867	0.486	4.9
YOLOv8n (Proposed)	3.0M	0.871	0.491	4.2
YOLOv8s	11.1M	0.860	0.480	6.9

Table 8 presents a comparison of three YOLO variants evaluated on the independent test set. YOLOv8n achieves the highest overall mAP@0.5 of 87.1% with the lowest inference latency of 4.2ms, while maintaining a compact model size of 3.0M parameters. YOLOv8s, despite having 3.7x more parameters and 3.5x higher computational cost, achieves a lower mAP@0.5 of 86.0%, suggesting that the larger model was not fully optimized within the available training epochs. YOLOv5n performs competitively at 86.7% mAP@0.5 with the smallest footprint of 2.5M parameters, making it a viable alternative for highly resource-constrained deployments. Based on the balance of accuracy, speed, and model size, YOLOv8n was selected as the proposed system's detection backbone.

Recent implementations, such as those by Ben Atitallah et al. [4], demonstrate the use of optimized YOLOv5 models via pruning and quantization for efficient embedded deployment. Similarly, Yu et al. [33] introduced LYTNNet, a real-time convolutional neural network for detecting pedestrian signals and zebra crossings.

Advanced algorithms like Dense Optical Flow (e.g., Farneback), DeepSORT and ByteTrack offer robust multi object tracking and dynamic motion awareness to motion analysis and tracking. In classical tracking pipelines, the Kalman Filter and the Hungarian Algorithm remain foundational components for object association and temporal tracking. Nevertheless, most contemporary systems fail to explicitly address ego-motion compensation, a gap that the proposed system resolves through homography-based correction and parallax adjustment.

The depth estimation can be divided into monocular techniques using deep learning networks like MiDaS or Monodepth2, or stereo vision systems and sensor solutions that use LiDAR or ultrasonic range-finders. For instance, Thakurdesai et al. [29] implemented monocular depth estimation alongside YOLO to deliver obstacle distance information. Kuriakose et al. [20] proposed an indoor navigation framework using deep learning on smartphone images to interpret spatial cues and signage for turn-by-turn guidance.

To convey processed information to users, feedback mechanisms employ auditory cues ranging from text-to-speech alerts to spatialized audio and haptic signals delivered through wearable devices, often combined in multimodal frameworks. Kuriakose et al. [20] developed DeepNAVI, a smartphone-based system providing scene context, obstacle type, and motion state through real-time audio. Furthermore, adaptive feedback strategies that adjust to environmental complexity and user context are emerging to optimize

usability. These existing technologies, combined, form an integrated ecosystem that supports enhanced navigation and obstacle avoidance for visually impaired users.

VI. DISCUSSION

The proposed system directly addresses three fundamental limitations common across prior assistive navigation systems. First, systems achieving motion awareness have typically relied on additional hardware — LiDAR [6], [22], depth cameras [19], or specialized headsets [25] — increasing cost and limiting portability. The proposed system achieves comparable motion awareness from a single monocular camera by treating depth estimation and ego-motion compensation as computational problems rather than hardware problems.

Second, prior systems using optical flow for motion detection [10] do not account for the depth-dependent reliability of motion cues. A vehicle extremely close to the camera produces large apparent optical flow vectors primarily due to camera motion, making flow-based classification unreliable at close range. Conversely, trajectory analysis requires sufficient temporal history and is unreliable for newly detected distant objects. The depth-adaptive weighting framework addresses this by dynamically prioritizing the most reliable cue at each depth range — a design approach not present in prior assistive navigation literature.

Third, prior systems do not encode the asymmetric safety cost of motion classification errors. The asymmetric hysteresis thresholds in the temporal consistency filter reflect the reality that in assistive navigation, false negative moving vehicle detections are significantly more dangerous than false positive alerts. This safety logic is encoded architecturally rather than as a post-processing step.

In general, it can be concluded that the identified limitations demonstrate the necessity of lightweight, computationally efficient, and motion-aware, real-time, and object detection assistive systems with feedback. Our suggested system is a direct solution to these difficulties, integrating effective YOLO-based detection and dense optical flow motion compensation, and is designed to be portable and responsive upon deployment on embedded hardware.

Although our proposed solution will successfully solve most of the fundamental issues that the current assistive navigation systems are dealing with, it has limitations as well. Quantitative evaluation under simulated low-light conditions revealed an overall mAP@0.5 of 0.818 compared to 0.871 under normal conditions, representing a 5.3% degradation as shown in Figure 5. Signal detection was most affected, dropping from 88.5% to 69.7%, attributed to the small bounding box size of traffic signals and reduced visual contrast under low illumination. Interestingly, obstacle and pedestrian detection showed marginal improvement under low-light conditions, likely due to increased contrast between subjects and background. Both the obstacle detection and motion awareness modules rely heavily on visual input, which can be significantly degraded under such conditions, leading to missed detections or false positives.

Moreover, there are tricky and highly dynamic environments where the movement detection component can sometimes be confused, wrongly classifying the motions of objects or not always tracking multiple motions occurring at the same time, e.g. a crowd of people. This has the potential of producing ambiguous or slow feedback to the user, which may decrease the safety and effectiveness in navigation. These difficulties highlight how hard real-time motion perception in cluttered scenes is, particularly when there are limited computing resources.

It is in light of these weaknesses that future work will aim at providing more robustness through the integration of improved algorithms. Additionally, quantitative evaluation of the motion classification module — specifically precision and recall for static versus moving vehicle discrimination — is identified as a current limitation and will be reported in a follow-up study with dedicated labeled video sequences.

A. ACCESSIBILITY IMPACT

The proposed system is effective in terms of solving major mobility problems under consideration of people with visual impairment since they provide real time knowledge of the dynamic surroundings as opposed to the traditional aids such as white canes or guide dogs. The system will allow users to move more confidently through crowded areas because it can utilize smart object recognition, motion tracking, and sound effects.

Nevertheless, to be really accessible, the system should be able to serve the various needs of its users. The visually impaired can differ a great deal with respect to auditory perception, speed of cognitive processing and reaction behavior. Subsequently, in the future versions it will be possible to customize the feedback parameters, such as speech rate, volume, frequency of repetitions and language support. These characteristics would enable the users to customize the system to an individual level of comfort and understanding to enhance the usability.

Besides, the hardware must be portable and easy to use in order to be deployable practically. Future iterations will focus on validating the system on lightweight, inexpensive embedded hardware to ensure accessibility across diverse user contexts., given that the user diversity and practical constraints, the system can serve more effectively in achieving its purpose of facilitating inclusive access to assistive technologies.

VII. CONCLUSION

This paper presents a real-time vision-based navigation assistance system that achieves ego-motion compensated dynamic obstacle classification exclusively from a single monocular camera, without LiDAR, stereo vision, depth sensors, or specialized wearable hardware. The primary novel contribution is a depth-adaptive multi-cue confidence fusion framework that dynamically reweights six complementary motion indicators based on estimated object depth, addressing the fundamental depth-dependent reliability of individual

motion cues in monocular vision. The system further incorporates asymmetric hysteresis thresholds encoding the safety-asymmetric cost of motion classification errors in assistive navigation. The proposed model achieves an overall mAP@0.5 of 87.1% on an independent held-out test set, maintaining 81.8% under simulated low-light conditions with an average pipeline latency of 46.73ms.

The system represents a promising step toward accessible assistive technology, providing situational awareness, ego-motion compensation, and contextual audio feedback, with real-world deployment validation as the immediate next direction.

The proposed system not only aims to enhance mobility for visually impaired users but also lays the groundwork for future developments involving 3D spatial cues, sensor fusion, and temporal modeling.

VIII. FUTURE WORK

Several directions are identified to extend this work toward real-world deployment and broader applicability. First, the system will be ported to embedded hardware platforms such as the NVIDIA Jetson Nano and Raspberry Pi 5 with chest- or glasses-mounted cameras to obtain empirical latency measurements and validate performance under genuine outdoor conditions. Second, a structured user study with visually impaired participants will be conducted under full institutional ethics approval to evaluate navigation confidence, audio feedback clarity, cognitive load, and response time in uncontrolled environments. Third, spatial awareness will be enhanced by incorporating 3D depth estimation to improve foreground-background separation in cluttered scenes. Fourth, multimodal sensor fusion integrating LiDAR, infrared, and ultrasonic inputs alongside RGB will address performance degradation under low-light conditions that camera-only systems cannot reliably handle. Finally, temporal deep learning models will be investigated to capture motion trends across frames, reducing transient classification errors and improving predictive reliability. Together, these extensions are expected to substantially advance the system's environmental resilience and practical accessibility impact.

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